

Cybersecurity Risk Assessment for ASBank Finance Ltd

Prepared By : AbdulQudus Olamilekan AbdulHakeem

Project Overview:

This project simulates a cybersecurity risk assessment conducted for a mid-sized financial organization known as **ASBank Finance Limited**. The purpose of the assessment is to identify potential cybersecurity threats, evaluate organizational risks, and recommend mitigation strategies to improve the company's overall security posture.

The assessment focuses on key organizational assets such as employee systems, internal networks, databases, web applications, and communication platforms. The project demonstrates foundational Governance, Risk, and Compliance (GRC) practices commonly used by cybersecurity professionals and risk analysts.

Objectives:

The objectives of this project are:

- Identify critical organizational assets
 - Analyze cybersecurity threats and vulnerabilities
 - Evaluate risk impact and likelihood
 - Assign risk ratings based on severity
 - Recommend mitigation and security controls
 - Improve organizational cybersecurity awareness
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Organization Profile:

Company Name:

ASBank Finance Ltd

Industry:

Financial Services

Organization Size:

30 Employees

Key Assets

- Employee laptops and workstations.
 - Customer database.
 - Email communication system.
 - Internal network infrastructure.
 - Web application server.
 - Wireless network.
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Risk Assessment Methodology:

The risk assessment was conducted using a qualitative risk analysis approach. Risks were evaluated based on:

- Likelihood of occurrence.
- Potential business impact.
- Existing vulnerabilities.
- Recommended mitigation controls.

Risk ratings were categorized as:

- Low
 - Medium
 - High
 - Critical
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Risk Register:

Risk ID	Asset	Threat	Vulnerability	Likelihood	Impact	Risk Level	Mitigation Strategy
R1	Employee Laptops	Malware Infection	Unsafe file downloads	High	High	Critical	Deploy endpoint protection and conduct employee security awareness training.
R2	Email System	Phishing Attack	Lack of phishing awareness	High	Medium	High	Implement email filtering and phishing awareness training
R3	Customer Database	Data Breach	Weak password policies	Medium	High	High	Enforce strong passwords and Multi-Factor Authentication (MFA).
R4	Web Server	DDoS Attack	Insufficient traffic filtering	Medium	Medium	Medium	Configure firewall protection and DDoS mitigation controls.
R5	Internal Network	Insider Threat	Excessive user privileges	Low	High	Medium	Apply least privilege access control policies.
R6	Wireless Network	Unauthorized Access	Weak Wi-Fi password	Medium	Medium	Medium	Implement WPA3 encryption and strong password policies.

Recommended Security Controls

The following security controls are recommended to reduce identified cybersecurity risks:

Technical Controls:

- Endpoint Detection and Response (EDR).
- Firewall configuration and monitoring.
- Multi-Factor Authentication (MFA).
- Security Information and Event Management (SIEM) Tools like Splunk / Wazuh.
- Network segmentation.
- Secure Wi-Fi encryption.

Administrative Controls:

- Cybersecurity awareness training.
- Password management policies.
- Access control policies.
- Incident response procedures.
- Regular security audits.

Operational Controls:

- Continuous monitoring of network activity.
- Routine vulnerability assessments.
- Backup and disaster recovery procedures.
- Log monitoring and event analysis.

Risk Analysis Summary:

The assessment identified phishing attacks, malware infections, weak authentication mechanisms, and unauthorized access as major cybersecurity risks affecting **ASBank Finance Ltd.** The highest risk identified was malware infection on employee laptops due to unsafe file downloads and inadequate cybersecurity awareness. Customer database security was also identified as a high-priority concern because of the financial and reputational impact associated with a potential data breach.

Implementing the recommended mitigation strategies would significantly reduce organizational risk exposure and strengthen overall cybersecurity resilience.

Conclusion:

This cybersecurity risk assessment project demonstrates how organizations can identify, evaluate, and mitigate cybersecurity risks through structured Governance, Risk, and Compliance (GRC) practices.

The project highlights the importance of combining technical security controls with administrative and operational policies to protect organizational assets and sensitive customer information.

By implementing the recommended controls, **ASBank Finance Ltd** can improve its security posture, reduce cyber risk exposure, and strengthen its overall cybersecurity governance framework.

Skills Demonstrated

- Cybersecurity Risk Assessment.
- Governance, Risk, and Compliance (GRC).
- Risk Analysis and Mitigation.
- Security Documentation.
- Cybersecurity Governance.
- Security Control Recommendations.
- Professional Reporting.